

Information Retrieval

Lecture 5

Evaluation of Retrieval Effectiveness

Evaluation

To place information retrieval on a systematic basis, we need **repeatable** criteria to **evaluate** how **effective** a system is in **meeting the information needs of the user** of the system.

This proves to be very difficult with a human in the loop. It proves hard to define:

- the task that the human is attempting
- the criteria to measure success

Matching:

Recall and Precision

With **matching methods**, if information retrieval were perfect ...

Every hit would be relevant to the original query, and every relevant item in the body of information would be found.

Precision: fraction (or percentage) of the hits that are relevant, i.e., the extent to which the set of hits retrieved by a query satisfies the requirement that generated the query.

Recall: fraction (or percentage) of the relevant items that are found by the query, i.e., the extent to which the query found all the items that satisfy the requirement.

Recall and Precision with Exact Matching

Example

- Corpus of 10,000 documents, 50 on a specific topic
- Ideal search finds these 50 documents and reject all others
- Actual search identifies 25 documents; 20 are relevant but 5 were on other topics
- Precision: $20/25 = 0.8$ (80% of hits were relevant)
- Recall: $20/50 = 0.4$ (40% of relevant were found)

Measuring Precision and Recall

Precision is easy to measure:

- A knowledgeable person looks at each document that is identified and decides whether it is relevant.
- In the example, only the 25 documents that are found need to be examined.

Recall is difficult to measure:

- To know all relevant items, a knowledgeable person must go through the entire collection, looking at every object to decide if it fits the criteria.
- In the example, all 10,000 documents must be examined.

Ranking Methods

Precision and recall measure the results of a **single query** using a **specific search system** applied to a **specific set of documents**.

Matching methods:

Precision and recall are single numbers.

Ranking methods:

Precision and recall are functions of the rank order.

Evaluating Ranking: Recall and Precision

If information retrieval were perfect ...

Every document relevant to the original information need would be **ranked** above every other document.

With ranking, precision and recall are **functions of the rank** order.

Precision(n): fraction (or percentage) of the n most highly ranked documents that are relevant.

Recall(n): fraction (or percentage) of the relevant items that are in the n most highly ranked documents.

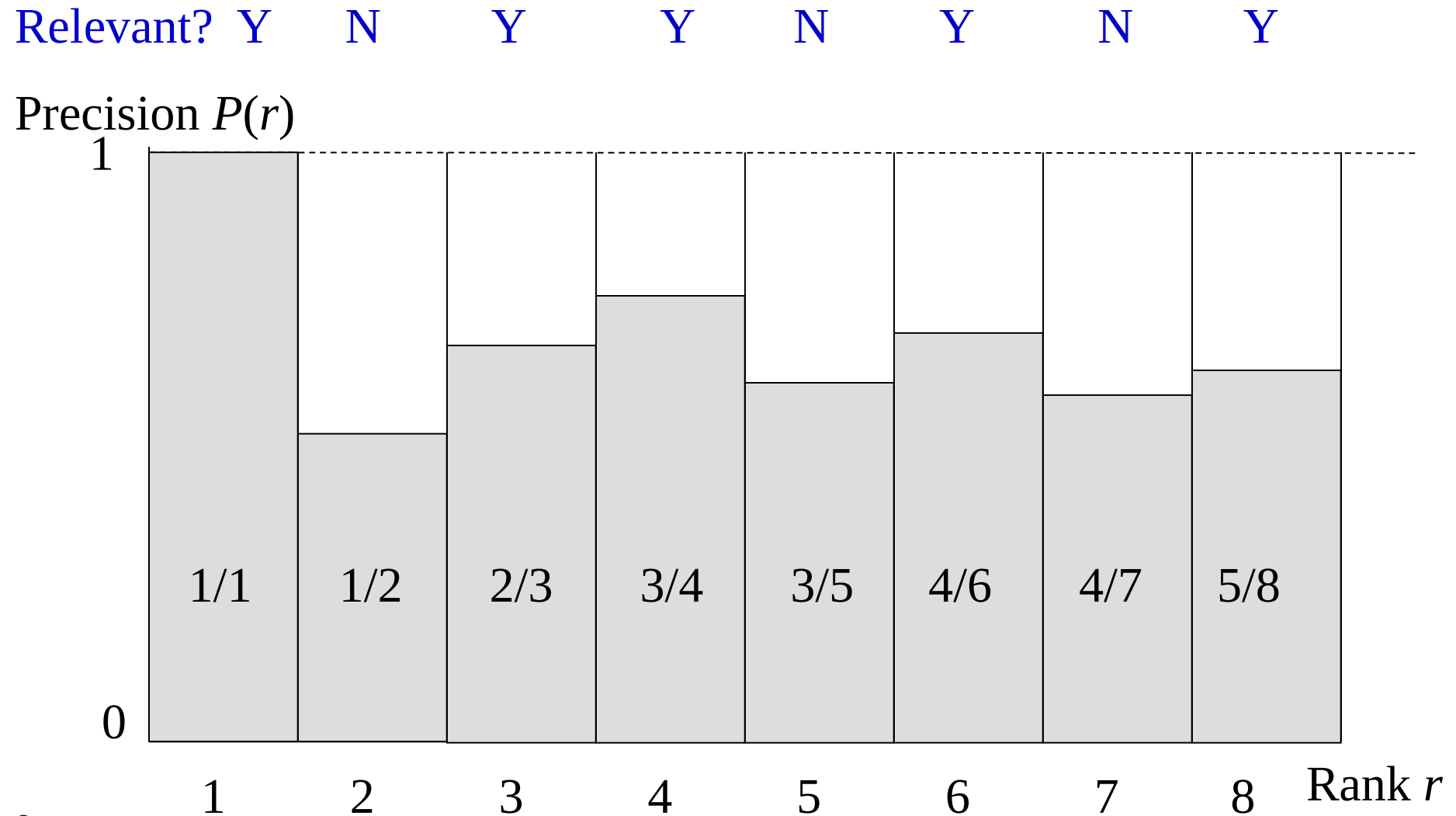
Precision and Recall with Ranking

Example

"Your query found 349,871 possibly relevant documents. Here are the first eight."

Examination of the first 8 finds that 5 of them are relevant.

Graph of Precision with Ranking: $P(r)$



Ranked retrieval: Recall and precision after retrieval of n documents

n	relevant	recall	precision
1	yes	0.2	1.0
2	yes	0.4	1.0
3	no	0.4	0.67
4	yes	0.6	0.75
5	no	0.6	0.60
6	yes	0.8	0.67
7	no	0.8	0.57
8	no	0.8	0.50
9	no	0.8	0.44
10	no	0.8	0.40
11	no	0.8	0.36
12	no	0.8	0.33
13	yes	1.0	0.38
14	no	1.0	0.36

Average precision

Average precision for a single topic is the mean of the precision obtained after each relevant document is obtained.

Mean average precision for a run consisting of many topics is the mean of the average precision scores for each individual topic in the run.

Relevance

Recall and precision: depend on concept of relevance

Relevance is a context-, task-dependent property of documents

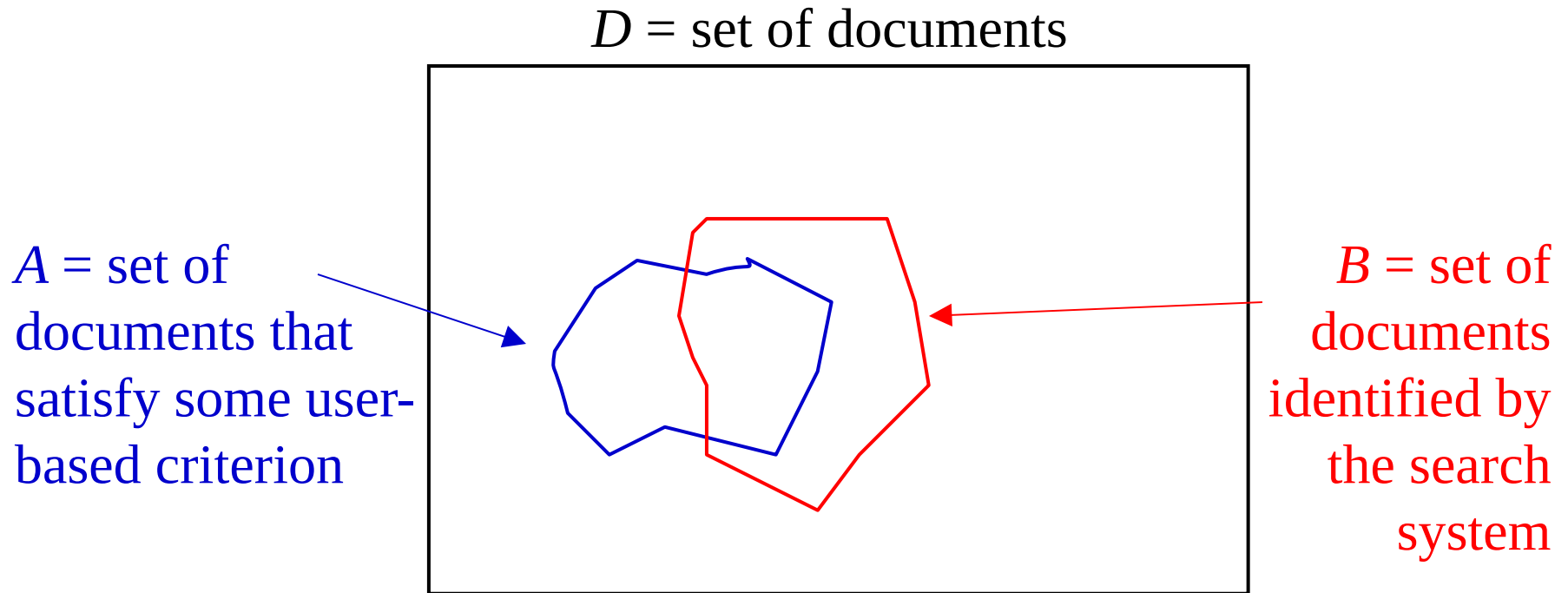
"Relevance is the correspondence **in context** between an **information requirement** statement ... and an article (a **document**), that is, the extent to which the article covers the material that is appropriate to the requirement statement."

Relevance

How stable are relevance judgments?

- For textual documents, knowledgeable users have good agreement in deciding whether a document is relevant to an information requirement.
- There is less consistency with non-textual documents, e.g., a photograph.
- Attempts to have users give a level of relevance, e.g., on a five point scale, are inconsistent.

Relevance as a Set Comparison



Measures based on relevance

$$\text{recall} = \frac{\text{retrieved relevant}}{\text{relevant}} = \frac{|A \cap B|}{|A|}$$

$$\text{precision} = \frac{\text{retrieved relevant}}{\text{retrieved}} = \frac{|A \cap B|}{|B|}$$

$$\text{fallout} = \frac{\text{retrieved not-relevant}}{\text{not-relevant}} = \frac{|B - A|}{|D - A|}$$